

Directions for Use

B. Braun Melsungen AG, Carl-Braun-Straße 1, 34212 Melsungen, Germany



Aminoplasma[®] Hepa – 10 %

Composition

1000 ml of solution contain

Active ingredients:

Isoleucine	8.80 g
Leucine	13.60 g
Lysine Acetate (equivalent to Lysine, 7.51 g)	10.60 g
Methionine	1.20 g
Phenylalanine	1.60 g
Threonine	4.60 g
Tryptophan	1.50 g
Valine	10.60 g
Arginine	8.80 g
Histidine	4.70 g
Glycine (Aminoacetic Acid)	6.30 g
Alanine	8.30 g
Proline	7.10 g
Aspartic Acid	2.50 g
Asparagine Monohydrate (equivalent to Asparagine, 0.48 g)	0.55 g
Acetylcysteine (equivalent to Cysteine, 0.59 g)	0.80 g
Glutamic Acid	5.70 g
Ornithine Hydrochloride (equivalent to Ornithine, 1.30 g)	1.66 g
Serine	3.70 g
Acetyltyrosine (equivalent to Tyrosine, 0.70 g)	0.86 g

Excipients:

Disodium edetate dihydrate	0.05 g
Hydrochloric acid or sodium hydroxide for pH adjustment, water for injections.	
Total amino acids	100 g/l
Total nitrogen	15.3 g/l
α-amino nitrogen	11.2 g
Caloric value	1675 kJ/l = 400 kcal/l
Osmolarity	875 mOsm/l

Electrolyte concentrations:

Acetate	51 mmol/l
Chloride	10 mmol/l

Product Description

A clear and colourless or faintly yellow solution.

Pharmaceutical form

Solution for infusion in glass bottles

Contents: 500 ml

Pharmaco-therapeutic group

Solution for supply of amino acids

Indications

Supply of amino acids in the setting of parenteral nutrition of patients with severe liver insufficiency and imminent or manifest hepatic encephalopathy.

Contraindications

- Disorders of amino acid metabolism of other than hepatic origin,
- Severely impaired circulatory status with vital threat (shock),
- Acute pulmonary oedema,
- Acidosis,
- Hyperhydration,
- Hypokalaemia,
- Hyponatraemia,
- Hypersensitivity to any of the ingredients of Aminoplasma[®] Hepa – 10 %

No data are available on the use of Aminoplasma[®] Hepa – 10 % in children under 2 years. Therefore, administration of the solution to such patients cannot be recommended until relevant data become available. Because of the particulars of its composition, Aminoplasma[®] Hepa – 10 % may cause marked metabolic disturbances if given for other reasons than stated under "Indications". Unindicated use must strictly be avoided.

Special warnings and precautions for use

Because of its composition, Aminoplasma[®] Hepa – 10 % should be administered to patients with concomitant renal insufficiency only after individual benefit/risk assessment.

The dose should be adjusted according to the serum urea and creatinine concentrations.

Caution is also to be exercised in patients with increased serum osmolarity.

Do not administer via peripheral venous catheter.

Amino acid therapy is not a substitute for established therapeutic measures, such as purging, administration of lactulose and/or gut sterilising antibiotics, in the treatment of hepatic encephalopathy.

Infusion of Aminoplasma[®] Hepa – 10 % should be accompanied by appropriate carbohydrate supply.

Electrolytes are to be supplemented according to actual requirements.

During parenteral therapy fluid and electrolyte balance, serum osmolarity, acid-base balance, blood glucose and liver function should be monitored. Type and frequency of the examinations depend on the severity of the patient's disease and clinical condition.

In particular, regular and more frequent clinical examinations and laboratory tests are necessary in patients with disorders of amino acid metabolism.

Special precautions for use in paediatrics:

The dosage must be adjusted according to the patient's age, nutritional status and the prevailing disease. In the case of supplementary or partial parenteral nutrition further protein supplying nutrients may have to be given.

An infusion from one bottle should not be carried out over longer than 24 hours.

For complete parenteral nutrition, administration of carbohydrates, essential fatty acids, vitamins and trace elements is necessary.

Pregnancy and lactation

No data are available on the use of Aminoplasma[®] Hepa – 10 % in these situations. It is likely that the conditions in which the product is indicated exclude pregnancy or breast-feeding.

Interactions

Because of the risk of microbial contamination and physico-chemical incompatibility, it is not recommended that any additives should be incorporated into Aminoplasma[®] Hepa – 10 % solution, but should preferably be given in standard carbohydrate or electrolyte solutions.

However, if admixture with Aminoplasma[®] Hepa – 10 % is essential then the compatibility of the additive must be checked before administration.

Dosage

Depending on individual requirements:

Normal dose: 7 – 10 ml /kg BW/day, corresponding to 0.7 – 1.0 g of amino acids/kg BW/day.

Maximum dose: 15 ml /kg BW/day, corresponding to 1.5 g of amino acids/kg BW/day.

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Flow rate:

Treatment of hepatic coma

In hepatic encephalopathy, it is recommended to start the infusion of Aminoplasma® Hepa – 10 % at an increased rate, until onset of the effect, for example for a patient weighing 70 kg:

1st to 2nd hour: 150 ml/h (2 ml/kg BW/h), corresponding to approx. 50 drops/min,
3rd to 4th hour: 75 ml/h (1 ml/kg BW/h), corresponding to approx. 25 drops/min,
from 5th hour: 45 ml/h (0.6 ml/kg BW/h), corresponding to approx. 15 drops/min.

Maintenance requirements/parenteral nutrition

45 – 75 ml/h, corresponding to 15 – 25 drops/min
(0.6 – 1.0 ml/kg BW/h)

Duration of use:

Aminoplasma® Hepa – 10 % may be administered as long as there is a risk of manifestation of hepatic encephalopathy

Method of administration

Intravenous infusion into the *vena cava*

Overdose

Symptoms

Overdosage or a too rapid infusion rate may manifest in the form of nausea, shivering, vomiting, and renal amino acid losses.

Emergency treatment, antidotes

In such cases the infusion should be interrupted and later continued at a lower infusion rate.

Undesirable effects

Provided contraindications, dosage recommendations and precautions are observed, side effects are not to be expected.

Note:

Patients should inform their doctor or pharmacist if they notice any adverse effect in connection with the administration of this medicine.

Pharmacodynamic properties

By administration of Aminoplasma® Hepa – 10 % with its amino acid composition specifically adapted to the pathologically altered metabolism of patients with liver cirrhosis, normalisation of amino acid imbalances can be achieved. Thus the cerebral manifestations of the disease, i.e. hepatic encephalopathy or hepatic precoma or coma are relieved and protein tolerance and protein biosynthesis are markedly improved.

Pharmacokinetic properties

The amino acid pattern of Aminoplasma® Hepa – 10 % is based on the results of investigations on the pharmacokinetics of amino acids in patients with liver cirrhosis. In those patients typical amino acid imbalances are present which are characterised by low concentrations of branched chain amino acids, increased concentrations of aromatic amino acids and increased concentration of methionine. These imbalances are regarded to be one of the causes of reduced protein tolerance and development of hepatic coma in patients with liver cirrhosis. The composition of Aminoplasma® Hepa – 10 %, characterised by a relatively high proportion of branched chain amino acids, has been adapted to the amino acid and protein metabolism of patients with liver cirrhosis. By administration of Aminoplasma® Hepa – 10 %, correction of the pathological amino acid pattern can be achieved.

Expiry date

The product must not be used beyond the expiry date stated on the labelling.

Storage

Keep containers in the outer carton in order to protect the contents from light.

Do not store above 30°C.

Instructions for use

The product is supplied in single-use containers. Administer immediately after connecting the container to the giving set. Unused contents must be discarded and should not be stored for later use.

The solution should not be administered if it is not clear or if the container or its closure show visible signs of damage.

Date of last revision:

04.2016

Product registration holder:

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